**ANNEX 1 (part A)**

**Innovation action**

**NUMBER — 760528 — MoniTank**

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# 1.1. The project summary

| Project Number [1] | 760528 | Project Acronym [2] | MoniTank |
|---|---|---|---|

One form per project

General information

| | |
|---|---|
| Project title [3] | Underground Storage Tanks Risk Mitigation System for petrol fuel stations |
| Starting date [4] | The first day of the month after the signature by the Commission |
| Duration in months [5] | 30 |
| Call (part) identifier [6] | H2020-FTIPilot-2016-1 |
| Topic | FTIPilot-01-2016<br>Fast Track to Innovation Pilot |
| Fixed EC Keywords | Advanced manufacturing and processing |
| Free keywords | Condition, monitoring, underground storage tanks, Acoustic emission |

Abstract [7]

MoniTank offers a commercial continuous advanced structural health monitoring system (SHM) for underground storage tanks (USTs) based on acoustic emission and guided wave technology, risk based inspection techniques, IoT and advance signal processing and data acquisition system. The product will be able to identify structural integrity faults with 99% accuracy of detecting structural problems in USTs before they lead to rupture or failure. Consortium including INT NDT, IRIS, TWI, BIC and Flotek will avail the business opportunity provided by market of 132k potential customers (filling stations in EU) and foreseen maintenance services investments in storage systems, resulting in growing NDT and maintenance needs and lack of robust autonomous continous SHM systems for USTs in oil and gas production and distribution industries market at present by offering MoniTank product. Limitations of current monitoring systems performance, their non compliance with legislation/standards requirement of detection accuracy and reliance on experienced workforce for inspection and maintenance create a differentiation strategic advantage for successful market replication of MoniTank.
Our product will be able to provide continuous monitoring and will use intelligent algorithms to predict potential structural failure. The technologies integrated and finalised during this project will consist of: sensors; processing capabilities (MCU processor hardware); AE sensors, transmission data; batteries; and data wireless transmission and cloud computing, therefore presenting the target customers with a system for risk based inspection criterion for USTs. Our vision is to grow our business by €24million in gross sales with a return on investment of about 15:1 in 5 years post project commercialisation. With business growth, we will create 112 new jobs. It is our strong belief that the Fast Track to Innovation Pilot is the ideal financial instrument for us to accelerate commercialization of MoniTank.

# 1.2. List of Beneficiaries

| Project Number [1] | 760528 | Project Acronym [2] | MoniTank |
|---|---|---|---|

| List of Beneficiaries | | | | | |
|---|---|---|---|---|---|
| **No** | **Name** | **Short name** | **Country** | **Project entry date[8]** | **Project exit date** |
| 1 | INTEGRITY NDT MUHENDISLIK SANAYI VE TICARET LIMITED SIRKETI | INT NDT | Turkey | | |
| 2 | IRIS TECHNOLOGY SOLUTIONS, SOCIEDAD LIMITADA | IRIS | Spain | | |
| 3 | TWI LIMITED | TWI | United Kingdom | | |
| 4 | FLOTEKS PLASTIK SANAYI VE TICARET AS | Floteks | Turkey | | |
| 5 | BRUNEL UNIVERSITY LONDON | BUL | United Kingdom | | |

# 1.3. Workplan Tables - Detailed implementation

## *1.3.1. WT1 List of work packages*

| WP Number[9] | WP Title | Lead beneficiary[10] | Personmonths[11] | Start month[12] | End month[13] |
|---|---|---|---|---|---|
| WP1 | Scaling up of MoniTank system as per commercial requirement | 1 - INT NDT | | 1 | 18 |
| WP2 | Testing and Performance validation of commercial system | 4 - Floteks | | 13 | 22 |
| WP3 | Commercial Product finalisation, Packaging and Certification | 1 - INT NDT | | 19 | 29 |
| WP4 | Product manufacturing, procurement and related supply chain | 1 - INT NDT | | 21 | 30 |
| WP5 | Dissemination, Exploitation and Communication Activities | 1 - INT NDT | | 3 | 30 |
| WP6 | MoniTank Sales , Marketing and Distribution Network | 1 - INT NDT | | 3 | 29 |
| WP7 | Project Management and Team Coordination | 1 - INT NDT | | 1 | 30 |
| | | **Total** | | | |

## Gantt chart of the Project:

| MoniTank | Leaders | Project Months | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| **WP1 - Scaling up of Monitank system as per commercial requirement** | **INT NDT** | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| **T1.1** Scaling up MoniTank system hardware to commercial version | INT NDT | | | | | | | | | | | | | | | | | | **D1.1 & M1** | | | | | | | | | | | | |
| **T1.2** Optimisation of Software and GUI Interface | BUL | | | | | | | | | | | | | | | | | | D1.2 | | | | | | | | | | | | |
| **T1.3** Scaled up system trials | INT NDT | | | | | | | | | | | | | | | | | | D1.3 | | | | | | | | | | | | |
| **WP2 - Testing and Performance validation of commercial system** | **Floteks** | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| **T2.1** System testing in factory and industrial environment | Floteks | | | | | | | | | | | | | | | | | | | | D2.1 | | | | | | | | | | |
| **T2.2** Performance verification and system validation | TWI | | | | | | | | | | | | | | | | | | | | D2.2 | | | | | | | | | | |
| **T2.3** Optimisation and customization of complete commercial product | INT NDT | | | | | | | | | | | | | | | | | | | | | | **M2** | | | | | | | | |
| **WP3 - Commercial Product finalisation, Packaging and Certification** | **INT NDT** | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| **T3.1** Finalisation of MoniTank's components and packaging | IRIS | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| **T3.2** Final Commercial market ready MoniTank Product | INT NDT | | | | | | | | | | | | | | | | | | | | | | | | | | | | **D3.1 & M3** | | |
| **T3.3** MoniTank certification, standards compliance and training development | TWI | | | | | | | | | | | | | | | | | | | | | | | | | | | | | D3.2 | |
| **WP4 - Product manufacturing, procurement and related supply chain** | **INT NDT** | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| **T4.1** MoniTank Product manufacturing and components procurement plan | IRIS | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | **D.4.1 & M4** |
| **T4.2** System supply chain finalisation and related agreements | INT NDT | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| **WP5 - Dissemination, Exploitation and Communication Activities** | **INT NDT** | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| **T5.1** Dissemination and exploitation plan and IP management | INT NDT | | | D5.1 | | | | | | | | | | | | | | | | | | | | | | | | D5.2 | | | **M5** |
| **T5.2** Communication and Advertising including demos development | IRIS | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| **T5.3** Awareness Events and Promotion | BUL | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| **T5.4** MoniTank pre-launch activities and clients engagement | INT NDT | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | D5.3 |
| **WP6 - Monitank Sales, Marketing and Distribution Network** | **INT NDT** | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| **T6.1** Monitank sales and marketing plan | INT NDT | | | | | | | | | | | | | | | | | | | | | | | | | | | | | **D6.1 & M6** | |
| **T6.2** Distribution sales agents network establishment | INT NDT | | | | | | | | | | | | | | | | | | | | | | | | | | | | | D6.2 | |
| **T6.3** After sales support services and maintenance procedures development | IRIS | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| **T6.4** Business development and product support team training | INT NDT | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| **WP7 - Product management and Team Coordination** | **INT NDT** | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| **T7.1** Financial and administrative management | INT NDT | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| **T7.2** Risk and innovation management | BUL | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| **T7.3** Product development Reporting and progress meetings | INT NDT | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| **T7.4** Quality assurance and Performance monitoring | INT NDT | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| **First year report** | **INT NDT** | | | | | | | | | | | | | | | | | | **D7.1** | | | | | | | | | | | | |
| **Project final report** | **INT NDT** | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | **D7.2** |
| **Risk management report** | **BUL** | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | **D7.3** |

### *1.3.2. WT2 list of deliverables*

| Deliverable Number[14] | Deliverable Title | WP number[9] | Lead beneficiary | Type[15] | Dissemination level[16] | Due Date (in months)[17] |
|---|---|---|---|---|---|---|
| D1.1 | Final MoniTank system scaled up version | WP1 | 1 - INT NDT | Demonstrator | Confidential, only for members of the consortium (including the Commission Services) | 18 |
| D1.2 | User Friendly GUI | WP1 | 5 - BUL | Demonstrator | Confidential, only for members of the consortium (including the Commission Services) | 18 |
| D1.3 | Product Manual | WP1 | 1 - INT NDT | Report | Confidential, only for members of the consortium (including the Commission Services) | 18 |
| D2.1 | Report on installation of the MoniTank system in underground storage tanks | WP2 | 4 - Floteks | Report | Confidential, only for members of the consortium (including the Commission Services) | 20 |
| D2.2 | Report of MoniTank operation including performance assessment and final customization | WP2 | 3 - TWI | Report | Confidential, only for members of the consortium (including the Commission Services) | 20 |

| | | | | | | |
|---|---|---|---|---|---|---|
| D3.1 | Final Commercial market Ready MoniTank System | WP3 | 1 - INT NDT | Demonstrator | Confidential, only for members of the consortium (including the Commission Services) | 28 |
| D3.2 | Report on Certification ,standards and trainings | WP3 | 3 - TWI | Report | Confidential, only for members of the consortium (including the Commission Services) | 29 |
| D4.1 | MoniTank system procurement and supply chain report | WP4 | 2 - IRIS | Report | Public | 30 |

| **Deliverable Number[14]** | **Deliverable Title** | **WP number[9]** | **Lead beneficiary** | **Type[15]** | **Dissemination level[16]** | **Due Date (in months)[17]** |
|---|---|---|---|---|---|---|
| D5.1 | Project website | WP5 | 2 - IRIS | Websites, patents filling, etc. | Public | 3 |
| D5.2 | Plan for the Exploitation and Dissemination of Results (PEDR) | WP5 | 1 - INT NDT | Report | Confidential, only for members of the consortium (including the Commission Services) | 27 |
| D5.3 | MoniTank Pre-Launch and promotion activities report | WP5 | 1 - INT NDT | Report | Confidential, only for members of the consortium (including the Commission Services) | 30 |
| D6.1 | MoniTank sales and marketing plan report | WP6 | 1 - INT NDT | Report | Confidential, only for members of the consortium (including the Commission Services) | 29 |
| D6.2 | Report on distribution sales network, After sales support services, product training and maintenance | WP6 | 1 - INT NDT | Report | Confidential, only for members of the consortium (including the Commission Services) | 29 |

| | | | | | | |
|---|---|---|---|---|---|---|
| D7.1 | First year final report | WP7 | 1 - INT NDT | Report | Confidential, only for members of the consortium (including the Commission Services) | 18 |
| D7.2 | Project final report | WP7 | 1 - INT NDT | Report | Confidential, only for members of the consortium (including the Commission Services) | 30 |
| D7.3 | Risk Management Report | WP7 | 5 - BUL | Report | Confidential, only for members of the consortium (including the Commission Services) | 30 |

### *1.3.3. WT3 Work package descriptions*

| **Work package number** [9] | WP1 | **Lead beneficiary** [10] | 1 - INT NDT |
|---|---|---|---|
| **Work package title** | Scaling up of MoniTank system as per commercial requirement | | |
| **Start month** | 1 | **End month** | 18 |

**Objectives**

To scale up the hardware and software of MoniTank System to commercial version.

**Description of work and role of partners**

**WP1 - Scaling up of MoniTank system as per commercial requirement** [Months: 1-18]
**INT NDT**, IRIS, TWI, Floteks, BUL
Task 1.1 Scaling up MoniTank system hardware to commercial version(Leader: INT NDT, Contributor: IRIS, Flotek, TWI,BUL)
MoniTank system will be scaled up as per commercial requirement provided by FLOTEKS. Size, cost, manufacturability and operational requirements will be considered in scaling up components. The hardware scale up of the acoustic emission system and acoustic emission sensor network will take into account mass production, such as supplier location and cost of transport etc. Further, risk based assessment analysis algorithm will be fine-tuned by TWI for the accuracy during use in the operational environment to be maintained at the same levels achieved until now. Data acquisition system and signal processing unit will be optimised in this task as well by BUL.

| **Deliverable Number**[14] | **Deliverable Title** | **Lead beneficiary** | **Type**[15] | **Dissemination level**[16] | **Due Date (in months)**[17] |
|---|---|---|---|---|---|

| | | | | | |
|---|---|---|---|---|---|
| D1.1 | Final MoniTank system scaled up version | 1 - INT NDT | Demonstrator | Confidential, only for members of the consortium (including | 18 |

Task 1.2 Optimisation of Software and GUI Interface(Leader: BUL, Contributor: IRIS)
MoniTank integrated software for automatic fault recognition will be optimised for scaled up system , a GUI will enable users to operate the system for monitoring and inspection of underground pipes
Task 1.3 Scaled up system trials(Leader INT NDT, Contributor: Floteks, IRIS)
Scaled up hardware and software system of MoniTank will be tested in underground tanks of FLOTEK. The initial trials will be done for optimisation and fine tuning of hardware and software as well as data acquisition system as per commercial performance requirements.

| Participation per Partner | |
|---|---|
| **Partner number and short name** | **WP1 effort** |
| 1 - INT NDT | |
| 2 - IRIS | |
| 3 - TWI | |
| 4 - Floteks | |
| 5 - BUL | |
| **Total** | |

List of deliverables

## List of deliverables

| Deliverable Number[14] | Deliverable Title | Lead beneficiary | Type[15] | Dissemination level[16] | Due Date (in months)[17] |
|---|---|---|---|---|---|
| | | | | the Commission Services) | |
| D1.2 | User Friendly GUI | 5 - BUL | Demonstrator | Confidential, only for members of the consortium (including the Commission Services) | 18 |
| D1.3 | Product Manual | 1 - INT NDT | Report | Confidential, only for members of the consortium (including the Commission Services) | 18 |

## Description of deliverables

D1.1 : Final MoniTank system scaled up version [18]
Final MoniTank system scaled up version

D1.2 : User Friendly GUI [18]
User Friendly GUI

D1.3 : Product Manual [18] Product Manual

## Schedule of relevant Milestones

| Milestone number[18] | Milestone title | Lead beneficiary | Due Date (in months) | Means of verification |
|---|---|---|---|---|
| MS1 | Scaled up system trials | 1 - INT NDT | 18 | Scaled up system trials - Varified by Deliverable D1.1 |

## Objectives

To validate the commercial MoniTank system in industrial environment and finalise market ready MoniTank Product.

## Description of work and role of partners

**WP2 - Testing and Performance validation of commercial system** [Months: 13-22]
**Floteks**, INT NDT, IRIS, TWI, BUL
Task 2.1 System testing in industrial environment(Leader :Floteks, Contributor: INT NDT,IRIS)
The final version of our product, achieved in WP1, will have to be demonstrated in an industrially operational environment. MoniTank sensor systems with signal transmitters will be fitted on underground storage tanks. The systems will gather and analyse data for a period of up to two months regarding the underground storage tanks operation. During this time, the extracted conclusions about the tank's condition will be periodically reported to the base application to analyse risks and plan maintenance operations as per need and criticality.

| | | | |
|---|---|---|---|
| **Work package number** [9] | WP2 | **Lead beneficiary** [10] | 4 - Floteks |
| **Work package title** | Testing and Performance validation of commercial system | | |
| **Start month** | 13 | **End month** | 22 |

Task 2.2 Performance verification and system validation (Leader :TWI, Contributor: INT NDT, Flotek, IRIS) MoniTank system performance verification will be done during testing in relevant environment. System final validation will be done after testing in different kind of underground storage tanks under different storage condition.

Task 2.3 Optimisation and customization of complete commercial product(Leader :INT NDT Contributor: IRIS) Based on performance testing, final customization of complete MoniTank subcomponents will be done.

Participation per Partner

| **Partner number and short name** | **WP2 effort** |
|---|---|
| 1 - INT NDT | |
| 2 - IRIS | |
| 3 - TWI | |
| 4 - Floteks | |
| 5 - BUL | |
| **Total** | |

List of deliverables

| **Deliverable Number**[14] | **Deliverable Title** | **Lead beneficiary** | **Type**[15] | **Dissemination level**[16] | **Due Date (in months)**[17] |
|---|---|---|---|---|---|
| D2.1 | Report on installation of the MoniTank system in underground storage tanks | 4 - Floteks | Report | Confidential, only for members of the consortium (including the Commission Services) | 20 |
| D2.2 | Report of MoniTank operation including performance assessment and final customization | 3 - TWI | Report | Confidential, only for members of the consortium (including | 20 |
| **Deliverable Number**[14] | **Deliverable Title** | **Lead beneficiary** | **Type**[15] | **Dissemination level**[16] | **Due Date (in months)**[17] |
| | | | | the Commission Services) | |

Description of deliverables

D2.1 : Report on installation of the MoniTank system in underground storage tanks [20]

Report on installation of the MoniTank system in underground storage tanks

## List of deliverables

D2.2 : Report of MoniTank operation including performance assessment and final customization [20]
Report of MoniTank operation including performance assessment and final customization

## Schedule of relevant Milestones

| **Milestone number**[18] | **Milestone title** | **Lead beneficiary** | **Due Date (in months)** | **Means of verification** |
|---|---|---|---|---|
| MS2 | Optimisation and customization of complete commercial product | 4 - Floteks | 22 | Optimisation and customization of complete commercial product - Varified by Deliverable D2.2 |

| | | | |
|---|---|---|---|
| **Work package number** [9] | WP3 | **Lead beneficiary** [10] | 1 - INT NDT |
| **Work package title** | Commercial Product finalisation, Packaging and Certification | | |
| **Start month** | 19 | **End month** | 29 |

**Objectives**

To finalise commercial market ready version of MoniTank Product.

**Description of work and role of partners**

**WP3 - Commercial Product finalisation, Packaging and Certification** [Months: 19-29]
**INT NDT**, IRIS, TWI, Floteks, BUL
Task 3.1 Finalisation of MoniTank's components and packaging(Leader: IRIS, Contributor: INT NDT)
MoniTank system will be finalised including packaging. For packaging intrinsic safe electronic device packaging will be considered. Various materials (such as plastic or resin encapsulation) and techniques (such as injection moulding, vacuum casting or thermal forming) will be considered in order to balance the relevant trade-offs. MoniTank system components, procurement will be finalised in this task as well.
Task 3.2 Final Commercial market ready MoniTank Product (Leader: INT NDT, Contributor: TWI, IRIS, Flotek)
Market ready commercial MoniTank Product will be finalised in this tasks with all necessary packaging and complete system.

Task 3.3 MoniTank certification, standards compliance and training development(Leader: TWI, Contributor: BUL)
MoniTank certification will be done in this task including any governing standards compliance. Training of implementing MoniTank system will be developed for customers as well.

**Participation per Partner**

| **Partner number and short name** | **WP3 effort** |
|---|---|
| 1 - INT NDT | |
| 2 - IRIS | |
| 3 - TWI | |
| 4 - Floteks | |
| 5 - BUL | |
| **Total** | |

**List of deliverables**

| **Deliverable Number**[14] | **Deliverable Title** | **Lead beneficiary** | **Type**[15] | **Dissemination level**[16] | **Due Date (in months)**[17] |
|---|---|---|---|---|---|

## List of deliverables

| | | | | | |
|---|---|---|---|---|---|
| D3.1 | Final Commercial market Ready MoniTank System | 1 - INT NDT | Demonstrator | Confidential, only for members of the consortium (including the Commission Services) | 28 |
| D3.2 | Report on Certification ,standards and trainings | 3 - TWI | Report | Confidential, only for members of the consortium (including | 29 |
| **Deliverable Number[14]** | **Deliverable Title** | **Lead beneficiary** | **Type[15]** | **Dissemination level[16]** | **Due Date (in months)[17]** |
| | | | | the Commission Services) | |

## Description of deliverables

D3.1 : Final Commercial market Ready MoniTank System [28]

Final Commercial market Ready MoniTank System

D3.2 : Report on Certification ,standards and trainings [29]

Report on Certification standards and trainings

## Schedule of relevant Milestones

| **Milestone number[18]** | **Milestone title** | **Lead beneficiary** | **Due Date (in months)** | **Means of verification** |
|---|---|---|---|---|
| MS3 | Final Commercial market ready MoniTank Product | 1 - INT NDT | 28 | Final Commercial market ready MoniTank Product varified by Deliverable D3.1 |

| | | | |
|---|---|---|---|
| **Work package number** [9] | WP4 | **Lead beneficiary** [10] | 1 - INT NDT |
| **Work package title** | Product manufacturing, procurement and related supply chain | | |
| **Start month** | 21 | **End month** | 30 |

**Objectives**

To develop a manufacturing, components procurement and supply chain plan for MoniTank Product.

**Description of work and role of partners**

**WP4 - Product manufacturing, procurement and related supply chain** [Months: 21-30]
**INT NDT**, IRIS, TWI, Floteks, BUL
Task 4.1 MoniTank Product manufacturing and components procurement plan (Leader: IRIS, Contributor: INT NDT)
MoniTank's mass production plan with the aim of reducing manufacturing cost and quality components, manufacturing and components procurement plan will be finalised. Agreements with suppliers to provide AE sensors, components, wires, packaging will be sorted out for producing commercial scale system for sales.
Task 4.2 System supply chain finalisation and related agreements(Leader: INT NDT Contributor: IRIS)
MoniTank system components supply chain required for components procurement and mass production of systems will be finalised and related agreements will be put in place for keeping up with the demand of commercial MoniTank product and related supply.

**Participation per Partner**

| **Partner number and short name** | **WP4 effort** |
|---|---|
| 1 - INT NDT | |
| 2 - IRIS | |
| 3 - TWI | |
| 4 - Floteks | |
| 5 - BUL | |
| **Total** | |

**List of deliverables**

| **Deliverable Number**[14] | **Deliverable Title** | **Lead beneficiary** | **Type**[15] | **Dissemination level**[16] | **Due Date (in months)**[17] |
|---|---|---|---|---|---|
| D4.1 | MoniTank system procurement and supply chain report | 2 - IRIS | Report | Public | 30 |

**Description of deliverables**

## List of deliverables

| |
|---|
| D4.1 : MoniTank system procurement and supply chain report [30]<br>MoniTank system procurement and supply chain report |

| Milestone number[18] | Milestone title | Lead beneficiary | Due Date (in months) | Means of verification |
|---|---|---|---|---|
| MS4 | Production Scale up finalised, suppliers selected | 1 - INT NDT | 30 | Production Scale up finalised, suppliers selected - varified by Deliverable D4.1 |

## List of deliverables

| | | | |
|---|---|---|---|
| **Work package number** [9] | WP5 | **Lead beneficiary** [10] | 1 - INT NDT |
| **Work package title** | Dissemination, Exploitation and Communication Activities | | |
| **Start month** | 3 | **End month** | 30 |

## Objectives

Objectives
To disseminate and exploit MoniTank system developed during the previous work packages as widely as possible. The objectives of this work package are:
To develop dissemination and exploitation plan (PEDR)
To promote the developed technique and maintain a technology transfer program that ensures successful implementation and provides a marketing platform to engage potential clients before launch
To ensure a widespread use and awareness of the developed MoniTank system this will contribute to increased petrol stations and underground storage operations safety.

## Description of work and role of partners

**WP5 - Dissemination, Exploitation and Communication Activities** [Months: 3-30]
**INT NDT**, IRIS, TWI, Floteks, BUL
In this task project dissemination including possible attendance to relevant events and software exhibitions, engaging potential clients/audience, disseminating the project in different regions by consortium partners. IPR between consortium members regarding foregrounds and commercial exploitation will be managed in this task as well. A detailed plan of exploitation and dissemination will be created and implemented in this task.
Task 5.2 Communication and advertising including demos development(Leader: IRIS, Contributor : all partners) Project website will be created for communication (between partners in internal members area) and for project marketing to public and external stakeholders/clients. Periodic communication about project development and commercialisation phase will be done in press releases and relevant newsletters including EUs portals and publishing sites. MoniTank video demo will be developed as well to demonstrate system's features and applications. Branding and advertising campaign will be developed in this task as well.
Task 5.3 Awareness Events and Promotion(Leader: BUL, Contributor : all partners)
MoniTank consortium will organise awareness events and software workshops inviting interesting stakeholders to create awareness of the system, product demo videos, press releases and advertising material will be developed to do promotion for engaging potential clients.
Task 5.4 MoniTank pre-launch activities and clients engagement (Leader: INT NDT, Contributor : all partners) MoniTank pre-launch activities will include building demand by B2B meetings and have pre-orders from engaged clients. Affiliate marketer will be sorted to engage clients at this stage as well. Distributors and sales agents will be trained about the product before launch as well.

## Participation per Partner

| Partner number and short name | WP5 effort |
|---|---|
| 1 - INT NDT | |
| 2 - IRIS | |
| 3 - TWI | |
| 4 - Floteks | |
| 5 - BUL | |

| | | | | | |
|---|---|---|---|---|---|
| | | | **Total** | | |
| **Deliverable Number[14]** | **Deliverable Title** | **Lead beneficiary** | **Type[15]** | **Dissemination level[16]** | **Due Date (in months)[17]** |
| D5.1 | Project website | 2 - IRIS | Websites, patents filling, etc. | Public | 3 |
| D5.2 | Plan for the Exploitation and Dissemination of Results (PEDR) | 1 - INT NDT | Report | Confidential, only for members of the consortium (including the Commission Services) | 27 |
| D5.3 | MoniTank Pre-Launch and promotion activities report | 1 - INT NDT | Report | Confidential, only for members of the consortium (including the Commission Services) | 30 |

**Description of deliverables**

D5.1 : Project website [3]
Project website

D5.2 : Plan for the Exploitation and Dissemination of Results (PEDR) [27]
Plan for the Exploitation and Dissemination of Results (PEDR)

D5.3 : MoniTank Pre-Launch and promotion activities report [30] MoniTank Pre-Launch and promotion activities report

**Schedule of relevant Milestones**

| **Milestone number[18]** | **Milestone title** | **Lead beneficiary** | **Due Date (in months)** | **Means of verification** |
|---|---|---|---|---|
| MS5 | Dissemination and exploitation plan and IP management | 1 - INT NDT | 30 | Dissemination and exploitation plan and IP management - varified by D5.2 |

| | | | |
|---|---|---|---|
| **Work package number [9]** | WP6 | **Lead beneficiary [10]** | 1 - INT NDT |
| **Work package title** | MoniTank Sales , Marketing and Distribution Network | | |
| **Start month** | 3 | **End month** | 29 |

**Objectives**

To develop MoniTank sales, marketing plan and customers support procedures (pre and post sales)

**Description of work and role of partners**

**WP6 - MoniTank Sales , Marketing and Distribution Network** [Months: 3-29]
**INT NDT**, IRIS, TWI, Floteks, BUL
Task 6.1 MoniTank sales and marketing plan(Leader: INT NDT, Contributor : all partners)
A detailed sales and marketing plan will be developed to be implemented during and post project commercialisation phase including detailed strategy of marketing and sales to potential clients in different regions.
Task 6.2 Distribution sales agents network establishment(Leader: INT NDT, Contributor : all partners)
Distribution sales agent network will be established in targeted regions before launch of the system to leverage maximum sales and customer approach.
Task 6.3 After sales support services and maintenance procedures development(Leader: IRIS, Contributor : all partners)
After sales support service will be developed in this task to be provided to customers of MoniTank including technical support, product enquiry and maintenance.

| **Deliverable Number**[14] | **Deliverable Title** | **Lead beneficiary** | **Type**[15] | **Dissemination level**[16] | **Due Date (in months)**[17] |
|---|---|---|---|---|---|
| D6.1 | MoniTank sales and marketing plan report | 1 - INT NDT | Report | Confidential, only for members of the consortium (including the Commission Services) | 29 |

| | | | | | |
|---|---|---|---|---|---|
| D6.2 | Report on distribution sales network, After sales | 1 - INT NDT | Report | Confidential, only for members of the consortium (including | 29 |

Task 6.4 Business development and product support team training(Leader: INT NDT, Contributor : all partners) Business development and product support team will be trained about the system, its functionality, features and value added benefit for customers to support sales and business development activities

Participation per Partner

| Partner number and short name | WP6 effort |
|---|---|
| 1 - INT NDT | |
| 2 - IRIS | |
| 3 - TWI | |
| 4 - Floteks | |
| 5 - BUL | |
| **Total** | |

List of deliverables

| Deliverable Number[14] | Deliverable Title | Lead beneficiary | Type[15] | Dissemination level[16] | Due Date (in months)[17] |
|---|---|---|---|---|---|
| | support services, product training and maintenance | | | the Commission Services) | |

Description of deliverables

D6.1 : MoniTank sales and marketing plan report [29]

MoniTank sales and marketing plan report

D6.2 : Report on distribution sales network, After sales support services, product training and maintenance [29]

Report on distribution sales network, After sales support services, product training and maintenance

Schedule of relevant Milestones

| Milestone number[18] | Milestone title | Lead beneficiary | Due Date (in months) | Means of verification |
|---|---|---|---|---|
| MS6 | MoniTank sales and marketing plan | 1 - INT NDT | 29 | MoniTank sales and marketing plan - varified by Deliverable D6.1 |

## List of deliverables

| | | | |
|---|---|---|---|
| **Work package number** [9] | WP7 | **Lead beneficiary** [10] | 1 - INT NDT |
| **Work package title** | Project Management and Team Coordinatior | | |
| **Start month** | 1 | **End month** | 30 |

## Objectives

To monitor and record progress and delivery of work packages.
To inform the Commission of progress made.(administrative and financial reporting)
To optimise the resources of the project team.
To manage risks and innovation aspects
To ensure best use is made of project resources; develop and maintain the project's website
To resolve any problems and disputes that may arise during the project and cannot be solved at individual work package level.

## Description of work and role of partners

**WP7 - Project Management and Team Coordination** [Months: 1-30]
**INT NDT**, IRIS, TWI, Floteks, BUL
Task 7.1 Financial and administrative management(Leader: INT NDT, Contributor : all partners)
The project coordinator will have responsibility for project files, ensuring correct and efficient knowledge management and for obtaining audit certificates from each consortium member as required. Coordinator will make sure that project funds are managed appropriately and will maintain project accounts. Finally, coordinator will ensure efficient communication and prompt reporting.
Task 7.2 Risk and innovation management (Leader: BUL, Contributor : all partners)
A risk and innovation management plan for both commercial and engineering development will be carried out throughout the project. The consortium will apply PFMEA (Process Failure Mode and Effects Analysis) and assign quality risk levels to each step in a process (manufacturing or transactional). The risks identified in the project will be continuously monitored.
Task 7.3 Product development Reporting and progress meetings(Leader: INT NDT, Contributor : all partners) Coordinator will manage project reporting including submission of deliverables, periodic progress reports and financial forms to the Commission. Periodic progress meetings between consortium members will be organised to track the progress of the project as well.
Task 7.4 Quality assurance and Performance monitoring (Leader: INT NDT, Contributor : all partners)
The coordinator will make sure about the quality of work being undertaken and will continuously monitor performance of project tasks and partners. Coordinator will make sure all milestones and deliverables of all WPs are up to standard and fulfil quality criteria established in the project.

## Participation per Partner

| **Partner number and short name** | **WP7 effort** |
|---|---|
| 1 - INT NDT | |
| 2 - IRIS | |
| 3 - TWI | |
| 4 - Floteks | |
| 5 - BUL | |
| **Total** | |

| Deliverable Number[14] | Deliverable Title | Lead beneficiary | Type[15] | Dissemination level[16] | Due Date (in months)[17] |
|---|---|---|---|---|---|
| D7.1 | First year final report | 1 - INT NDT | Report | Confidential, only for members of the consortium (including the Commission Services) | 18 |
| D7.2 | Project final report | 1 - INT NDT | Report | Confidential, only for members of the consortium (including the Commission Services) | 30 |
| D7.3 | Risk Management Report | 5 - BUL | Report | Confidential, only for members of the consortium (including the Commission Services) | 30 |

Description of deliverables

D7.1 : First year final report [18]
First year final report
D7.2 : Project final report [30]
Project final report
D7.3 : Risk Management Report [30]
Risk Management Report

Schedule of relevant Milestones

| Milestone number[18] | Milestone title | Lead beneficiary | Due Date (in months) | Means of verification |
|---|---|---|---|---|

### *1.3.4. WT4 List of milestones*

| Milestone number[18] | Milestone title | WP number[9] | Lead beneficiary | Due Date (in months)[17] | Means of verification |
|---|---|---|---|---|---|
| MS1 | Scaled up system trials | WP1 | 1 - INT NDT | 18 | Scaled up system trials - Varified by Deliverable D1.1 |
| MS2 | Optimisation and customization of complete commercial product | WP2 | 4 - Floteks | 22 | Optimisation and customization of complete commercial product - Varified by Deliverable D2.2 |

| | | | | | |
|---|---|---|---|---|---|
| MS3 | Final Commercial market ready MoniTank Product | WP3 | 1 - INT NDT | 28 | Final Commercial market ready MoniTank Product varified by Deliverable D3.1 |
| MS4 | Production Scale up finalised, suppliers selected | WP4 | 1 - INT NDT | 30 | Production Scale up finalised, suppliers selected - varified by Deliverable D4.1 |
| MS5 | Dissemination and exploitation plan and IP management | WP5 | 1 - INT NDT | 30 | Dissemination and exploitation plan and IP management - varified by D5.2 |
| MS6 | MoniTank sales and marketing plan | WP6 | 1 - INT NDT | 29 | MoniTank sales and marketing plan - varified by Deliverable D6.1 |

### *1.3.5. WT5 Critical Implementation risks and mitigation actions*

| Risk number | Description of risk | WP Number | Proposed risk-mitigation measures |
|---|---|---|---|
| 1 | MoniTank's RBI algorithms and scaled up hardware are incompatible. Risk: Medium | WP1 | Scaled up hardware components selection will explicitly take into account the algorithm computational requirements. TWI will perform thorough RBI analysis and software optimisation in order to keep computational load at a minimum. Underground storage tanks operators will provide inputs about compatibility of system and associated AE sensor network, and proper data acquisition. |
| 2 | MoniTank's accuracy in operational environment does not meet specs. Risk: Low | WP2 | RBI technique has been thoroughly benchmarked during trials in relevant environment. If necessary, more effort will be allocated in optimising the analysis algorithms quantitatively in order to match the accuracy levels. |
| 3 | High Manufacturing cost of combined system due to high priced supply components. Risk: Medium | WP4 | WP4 will be dedicated to production scale up and suppliers' selection and agreements, where price, availability and time of delivery of components will be agreed to make sure manufacturing cost don't vary as forecasted and projected for mass production and sales. |
| 4 | Certification approval of MoniTank system is not obtained. Risk: Low | WP3 | Consortium members have been already got in touch with the relevant authorities and their feedback is positive. Extra effort will be allocated for hardware and modifications if need be. |
| 5 | Market Initial Inertia in acceptance of innovative new technology Risk: Low | WP5 | Early engagement of potential customers will be done to make sure we have secured sales from important brand in fuel station retail market, considering SHELL,BP,TOTAL have already shown interest in buying commercial system on commercial validation, we are sure to capture significant market share with this new technological condition monitoring system. |
| 6 | Distribution Network is not established as planned. Risk: Low | WP6 | Coordinator's large client network and experience guarantees smooth progress with regards to distribution network establishment. The Commercial Manager will allocate more effort into leveraging his contacts to overcome any problems faced. |

### *1.3.6. WT6 Summary of project effort in person-months*

| | **WP1** | **WP2** | **WP3** | **WP4** | **WP5** | **WP6** | **WP7** | **Total Person/Months per Participant** |
|---|---|---|---|---|---|---|---|---|
| 1 - INT NDT | | | | | | | | |
| 2 - IRIS | | | | | | | | |
| 3 - TWI | | | | | | | | |
| 4 - Floteks | | | | | | | | |
| 5 - BUL | | | | | | | | |
| **Total Person/Months** | | | | | | | | |

## *1.3.7. WT7 Tentative schedule of project reviews*

| Review number [19] | Tentative timing | Planned venue of review | Comments, if any |
|---|---|---|---|
| RV1 | 12 | Remote | to be confirmed |
| RV2 | 24 | Remote | to be confirmed |



**1. Project number**

The project number has been assigned by the Commission as the unique identifier for your project. It cannot be changed. The project number **should appear on each page of the grant agreement preparation documents (part A and part B)** to prevent errors during its handling.

**2. Project acronym**

Use the project acronym as given in the submitted proposal. It can generally not be changed. The same acronym **should appear on each page of the grant agreement preparation documents (part A and part B)** to prevent errors during its handling.

**3. Project title**

Use the title (preferably no longer than 200 characters) as indicated in the submitted proposal. Minor corrections are possible if agreed during the preparation of the grant agreement.

**4. Starting date**

Unless a specific (fixed) starting date is duly justified and agreed upon during the preparation of the Grant Agreement, the project will start on the first day of the month following the entry into force of the Grant Agreement (NB : entry into force = signature by the Commission). Please note that if a fixed starting date is used, you will be required to provide a written justification.

**5. Duration**

Insert the duration of the project in full months.

**6. Call (part) identifier**

The Call (part) identifier is the reference number given in the call or part of the call you were addressing, as indicated in the publication of the call in the Official Journal of the European Union. You have to use the identifier given by the Commission in the letter inviting to prepare the grant agreement.

**7. Abstract 8. Project Entry Month**

The month at which the participant joined the consortium, month 1 marking the start date of the project, and all other start dates being relative to this start date.

**9. Work Package number**

Work package number: WP1, WP2, WP3, ..., WPn

**10. Lead beneficiary**

This must be one of the beneficiaries in the grant (not a third party) - Number of the beneficiary leading the work in this work package

**11. Person-months per work package**

The total number of person-months allocated to each work package.

**12. Start month**

Relative start date for the work in the specific work packages, month 1 marking the start date of the project, and all other start dates being relative to this start date.

**13. End month**

Relative end date, month 1 marking the start date of the project, and all end dates being relative to this start date.

**14. Deliverable number**

Deliverable numbers: D1 - Dn

**15. Type**

Please indicate the type of the deliverable using one of the following codes:

R Document, report
DEM Demonstrator, pilot, prototype DEC Websites, patent fillings, videos, etc.
OTHER
ETHICS Ethics requirement
ORDP Open Research Data Pilot

**16. Dissemination level**

Please indicate the dissemination level using one of the following codes:

PU Public
CO Confidential, only for members of the consortium (including the Commission Services)
EU-RES Classified Information: RESTREINT UE (Commission Decision 2005/444/EC)
EU-CON Classified Information: CONFIDENTIEL UE (Commission Decision 2005/444/EC) EU-SEC Classified Information: SECRET UE (Commission Decision 2005/444/EC)

**17. Delivery date for Deliverable**

Month in which the deliverables will be available, month 1 marking the start date of the project, and all delivery dates being relative to this start date.

**18. Milestone number**

Milestone number:MS1, MS2, ..., MSn

**19. Review number**

Review number: RV1, RV2, ..., RVn

**20. Installation Number**

Number progressively the installations of a same infrastructure. An installation is a part of an infrastructure that could be used independently from the rest.

**21. Installation country**

Code of the country where the installation is located or IO if the access provider (the beneficiary or linked third party) is an international organization, an ERIC or a similar legal entity.

**22. Type of access**

VA if virtual access,
TA-uc if trans-national access with access costs declared on the basis of unit cost,
TA-ac if trans-national access with access costs declared as actual costs, and
TA-cb if trans-national access with access costs declared as a combination of actual costs and costs on the basis of unit cost.

**23. Access costs**

Cost of the access provided under the project. For virtual access fill only the second column. For trans-national access fill one of the two columns or both according to the way access costs are declared. Trans-national access costs on the basis of unit cost will result from the unit cost by the quantity

## 4. Members of the Consortium

## 4.1. Participants

**Participant Number 1**

| | |
|---|---|
| INTEGRITY NDT | **Organisation Name: INTEGRITY NDT MUHENDISLIK SAN. TIC. VE LTD. (Integrity NDT) Website: www.integrityndt.com** |

**Profile:**
Integrity NDT Mühendislik (INDT) was formed in 2010 in Ankara, Turkey to provide NDT (non-destructive testing) services, inspection management services and asset management software to fabrication, power and process plant engineers.

Location and Contact
INTEGRITY NDT MÜHENDİSLİK SANAYİ VE TİCARET LİMİTED
ŞİRKETİ MelihGökcek Bulvarı EminelİşMerkezi
18/94 IV
OGSAN/ANKARA
Tel: (+90) 3123944331
Fax: 03123944332
www.integrityndt.com.tr

INDT is an SME with 11 full time employees.

INDT, certified to ISO 9001:2008, has become a key service provider and approved supplier to the likes of GDF Suez, TUPRAŞ (Turkish Petroleum Refineries), SABUL Refineries (KSA), Ipragaz LPG and PETKIM Refinery, delivering costeffective inspection management, advanced inspection application, and asset management and software solutions, combined with over 35 years' experienced engineers in planning and carrying out all aspects of NDT in fabrication, power and process plant worldwide.

INDT staff has a suite of independent inspection qualifications conforming to ISO 9712 standards Levels II and III in all major NDT methods including eddy current, Magnetic particle inspection, liquid penetrant inspection, radiography, ultrasonic inspection, weld inspection and metallurgical inspection. Additionally INDT specialise in automated ultrasonic testing including phased array (PAUT) and time of flight diffraction (TOFD) inspections and developing, verifying and certifying inspection solution applications.

INDT also provides commercial inspection management software tools; Clients in the power generation, oil & gas and petrochemical industries utilise INDT inspection systems. These systems are also applied in-house to contracts run by INDT, ensuring accurate QA/QC reporting and efficient project management.

INDT also have experience of selling in-house developed products such as our software systems, AUT system scanners and specialist application scanners and accessories.

Commercially available software systems include:-
**QUARTZ:** is a web based inspection management solution developed for planning and maintaining records over the pre and in- service life of a plant asset. The management of industrial plant is reliant on detailed system drawings, with unique identification assigned to critical items, for example, high pressure systems pipework; welds, hangers, and bends, etc. From available plant drawings and site surveys, Quartz links scaled 3D CAD pipework modelling, through 'live' on screen tags, to a bespoke database of inspection records and a document management system, with the ability to attach original construction related documentation, as well as records of associated ongoing inspections and system management documentation. Providing engineers with a powerful means for planning inspection regimes and result management, it facilitates the easy consolidation and analysis of reported inspection results and data.

**QUARTSITE:** We provide commercial outage management software 'QUARTSITE' into which we upload the
shutdown/outage programmed inspection activities. The software is designed specifically to manage and track individual task status, and control stored outage QA/QC requirements. Throughout the project duration, 'QUARTSITE' provides vital up-tothe-minute inspection status reports for every inputted activity, through to the final detailed test reports produced at project close-out.

The use of this software to control and manage work scope activities creates real efficiencies in the following areas; Defines the precise inspection work scope and the tests to be applied.

- Provides 'Work Pack' Instructions to Inspection Personnel, including specification of the procedures and techniques to be applied.
- Monitors and controls the progress and status of inspections, providing the client and other site teams with regular reports on progress, status and any items with commercial implications.
- Processes results through established acceptance criteria or through metallurgist or plant specialists
- Ensures that all activities are completed, the QA 'loop' is closed out, and the plant made available for service in the desired timescales
- Produces comprehensive controlled electronic inspection reports.

Mission and Objectives
INDT specialise in the inspection management of power plant outages and refinery shutdown 'turnarounds'. Through our industry wide experience of NDT we deliver a range of services in our approach to plant inspection and asset management, including a range of metallurgical analysis and materials testing. We also supply commercial inspection management software; it is the use of our in- house developed software systems which significantly adds value to our asset management approach to project management from outset to delivery.

A specialist provider of innovative inspection services and solutions, our quality is highly valued by ourselves and our client customers. From small local manufacturers to international companies worldwide, we are committed to delivering cost effective inspection solutions and technical assurance of the integrity of plant equipment and fabrications to our clients, regulatory and insurance bodies alike. Our quality systems are underpinned by ISO9001 certification.

Investing in our Engineering team is our investment in our future and what supports our integrity to our clients, and we are committed to a continual training programme. Our team of engineers supports our integrity and commitment to our clients. Across our team of engineers, INDT hold independent inspection qualifications conforming to ISO9712 standard, levels II and III in all methods; magnetic particle, liquid penetrant, eddy current, ultrasonic and radiographic inspection. Additionally INDT specialise in advanced ultrasonic testing TOFD and phased array. We have a clear focus is on the future and continual investment in technology in our field, in alignment with our commitment to our clients and their requirements is our priority.

Centred in Ankara Turkey we have; offices, workshops, metallurgy/development and training facilities.

**Role in Project:**
At the Phase 1 - Feasibility Study stage, INDT will be the single SME in the project and will act as Project Coordinator. The company directors/senior managers will make all of the decisions relating to the project as follows:- (a) All contact with the EC project officer/representative
(b) The day to day running of the project according to the work plan
(c) All dealings with prospective subcontractors
(d) Resolution of any problems that arise

**Reasons for Participation**:
INDT specialise in advanced NDT applications in the energy industry sector including oil &gas and petrochemical as well as power generation. They are therefore enthusiastic to explore the market potential for this specific technology application of LNG Tank inspection as they see significant EU and global market potential. Having significant experience in AUT on Tank inspections and

existing commercial links, are well positioned to commercialise and exploit the TANK-TEST system following Phase 2 engineering design, build, validation and demonstration should the Phase 1 Feasibility Study and Business Plan justify it.

**Existing Customers/End Users:**
GDF Suez
HUBCO HubPower Power Plant Energy Generation (Pakistan)
STANWELL Energy (Australia)
TUPRAŞ (Turkish Petroleum Refineries)
SABUL Chemical Manufacturing (KSA), Ipragaz LPG
(Turkey) PETKIM Refinery (Turkey)

**Key Personnel:**

**Görkem Bayten BSc Physics Engineering, MSc MBA.**
Görkem has been involved in NDT and Inspection Management for 8 years and holds a BSc Honours degree plus a suite of Level II qualifications in NDT and Weld Inspection,He specialises in Automated UT application including Phased Array, TOFD and the development of 'solutions' to specialist applications. Combined with extensive experience in devising and advising on asset management software solutions to plant operators.Görkem is also experienced in managing software development with regard to client requirements and the business development/marketing/distribution and systems management, for an existing and growing client base utilising Integrity NDT Software products.

**Özgen Bulut BSc Physics Engineering.**
Özgen has been involved in NDT and Inspection Management for 8 years and holds a BSc Honours degree plus a suite of Level II qualifications in NDT and Weld Inspection combined with extensive experience in devising and advising on asset management software solutions to plant operators. He also covers all other areas of the business, from outage inspection management to specialist ultrasonic testing applications such as TOFD, Phased Array, and High Frequency boiler tube Oxide Thickness measurements, and EMATs. He also worked on design, manufacture and sale of basic ultrasonic testing accessories such as probe splitters and umbilicals.

**Onur Yalim BSc Physics Engineering.**
Onur has been involved in NDT and Inspection Management for 2 years and holds a BSc Honours degree plus Level II qualifications in NDT and Weld Inspection. He is experienced in shutdown/outage inspection applications including AUT TOFD.

**John Stephenson IMechE, MInstNDT, LEP.**
John has been involved in NDT and Inspection Management for 30 years. He is a former employee of the Central Electricity Generating Board and Nuclear Electric and holds a suite of Level II and III qualifications in non-destructive testing. He specialises in outage or plant shutdown inspection management, and to date has managed inspection activities on over twenty Alstom 13E2 Gas Turbine and Steam Turbine shutdowns around the world. Additionally, John is highly experienced in carrying out semi and fully automated ultrasonic inspections and has designed and manufactured Integrity NDT semi-automated TOFD (Time Of Flight Diffraction) and Phased Array Scanner systems, as well as bespoke scanners for one-off special application solutions.
Software Systems - John has been involved in the design/development of Integrity NDT Software products for over 10 years, and has experience of marketing/sales and distribution to an existing client base.

**Participant Number**
**2**

| | |
|---|---|
| iris #tech4.0 | **Organisation Name: INNOVACIO I RECERCA INDUSTRIAL I SOSTENIBLE SL (IRIS)**<br>**Website:www.iris-eng.com** |

**Organisation Description:**

IRIS (www.iris.cat ) is a technology SME that was established in Barcelona in 2007 and specialises in photonics (spectroscopy) and sensor technology integration and SMAC (Social, Mobile, Analytics and Cloud) Technology for the development of advanced monitoring and control solutions and for IoT-enabling equipment and machinery for transitioning industry to the Industry4.0 paradigm. To date, our target sectors are the food industry, agriculture, pharmaceutical and chemicals.

We have engineering and production facilities in Spain and in 2016 we opened Commercial Offices in San Francisco and New York, as part of our international commercialization expansion strategy. We have a sister company in Dublin, VISUM Ltd. that specialises in the development of NIR spectroscopy-based measurement systems, and we also have a shareholding in a technology start-up in Finland that specialised in time-gated Raman (Timegate Instruments OY).

We are ISO 9001:2008 certified and we are one of the few Spanish companies to also have UNE-EN-ISO 166002: Management Systems R&D&I certification.

We currently employ 50+ staff and have a turnover of approx. € 4 M + IRIS is a member of the following networks: SPIE (International Society for Optics and Photonics), SECPHO (Southern European Cluster of Photonics), SPIRE (Sustainable Process Industries). IRIS is an SME Champion (has an SME Instrument award- Phase 2 project).

**Relevant Publications:**

- IRIS provides advanced monitoring solutions to the industry: www.iris.cat In terms of its market expertise, since 2011, IRIS has been building a portfolio of private B2B contracts and has developed advanced monitoring solutions and products for a number of companies, predominantly from Catalonia, but also serving some international companies, with references such as:GESCASER, TORRES, ESTRELLA DAMM, OLLER, ITPSA, ALMIRALL PRODESFARMA, NESTLE, ETC.
- E. Gaston (2013). Hyperspectral imaging (HSI) - an emerging platform technology that is set to revolutionise process monitoring and quality control. Pharmaprocess 2013, Innovation Forum in Pharmaceutical Process. Barcelona, 20 October 2013.
- E. Sola, C. Jumilla (2011). Composting monitoring: a real world challenge for WSNs, 4th IntelliCIS Workshop, Vilanovai la Geltru (Barcelona), Spain between 13-14 June 2011.
- "Wireless Sensor Network for Smart Composting Monitoring and Control".Oscar Casas, MargaLópez, Marcos Quílez, Xavier Martinez-Farre, Gemma Hornero, Carlos Rovira, Mirta R Pinilla, Pedro M Ramos, Beatriz Borges, Hugo Marques, Pedro Silva GirãoMeasurement 01/2014; 47:483 - 495.
- “Delivery of courses to industry in chemometrics tools for Process Analytical Technologies (PAT), e.g.: “Chemometric tools forprocess analytical technologies. ChemoPAT”, University of Tarragona 16-18 July 2014; PAT: Innovando en tecnología de controlde procesos farmacéuticos (Asociación de Químicos e Ingenieros del IQS)- May 28th 2015.

**Relevant Projects:**

**HYPERA:** Disruptive detection Technology Grant Agreement 726572, SME Instrument Phase II, on-going) IRIS Coordinator.

**IbD:** Intensified by design.Grant Agreement 680565. H2020 SPIRE, on-going) IRIS Coordinator.

**Propat:** Tackling the future of plant operation – How to integrate online data-analytics, process control, and PAT for optimal plant operation. Grant Agreement 637232. H2020 SPIRE, on-going) IRIS Coordinator.

**ShareBox:** Secure management platform for shared process resources (Grant agreement no.: 680843, H2020 SPIRE, on-going) IRIS Coordinator.

**WASTE2FUELS**. Sustainable production of next generation biofuels from waste streams. (Grant Agreement no: 654623 Horizon 2020-LCE-11, on-going). IRIS Coordinator

**Infrastructure/Equipment:**

IRIS is equipped with a Spectroscopic Laboratory equipped with several instruments, namely, an FTIR-ATR (Interspectrum), two
UVVIS(PerkinElmer and Ocean Optics), one Extended-SWIR (StellarNet) and one NIR (Mightex) spectrophotometers as well as a SWIRHyperspectral imager (Specim-Xenics). The associated staff are trained and skilled in chemometrics modelling by using a commercialsuite (SOLO+MIA from Eigenvector) as well as ad hoc non-commercial chemometric tools (Matlab and Python toolboxes). Electronics, mechanical and optical workshops and laboratories equipped with up-to-date hardware and tools: signal generators,wide band oscilloscopes, PCB design [Altium] and building tools, development kits for programming microcontrollers, CNC-millingmachines for electronic and mechanical prototyping, and CAD software [SolidWorks], among others. ZEEMAX software for the designof optical systems and design setup by means of advanced optical components and facilities (optical tables, windows, lenses, lasersand light sources and related “opticalware”).
In addition IRIS has an INNOVATION HUB with its own co-creation area that has floor-to-ceiling blackboards for brainstorming andconceptualisation, a demo and presentation area, a sprint room. In addition, IRIS has an auditorium for more formal presentations and training.

**Profile of Main Persons**

**MSc. ALEJANDRO ROSALES**. Male. M. Sc. in Physics from the Universidad de La Habana (1986). His background comprises theoretical and technical skills in several scientific and technological areas: optics, lasers, high vacuum, colorimetry, UV-VIS- NIRMIR spectroscopy, chemometrics and design of measurement instruments. He has extensive practical experience in programming, electronics and mechanical CAD design as well as R&D methodology and management. His career has included positions as researcher applications engineer, design engineer, business developer and R&D manager. In 2000s, he worked in the FTIR spectroscopy field applied to quality control in the wine industry, whereby he developed two new measurement instruments for wine analysis. His career as industrial physicist has been focused in applying his wide expertise to provide solutions for industrial problems. Knowledge in PAT, QbD, DoE and he has delivered numerous courses to industry in the field of practical PAT adoption. Knowledge in leading the building of chemometrics and multivariate analysis modelling tools. He is the Science and Technology Manager in IRIS. During this project, Alejandro will act as the Innovation Manager and he will also work closely with the engineering team in IRIS.

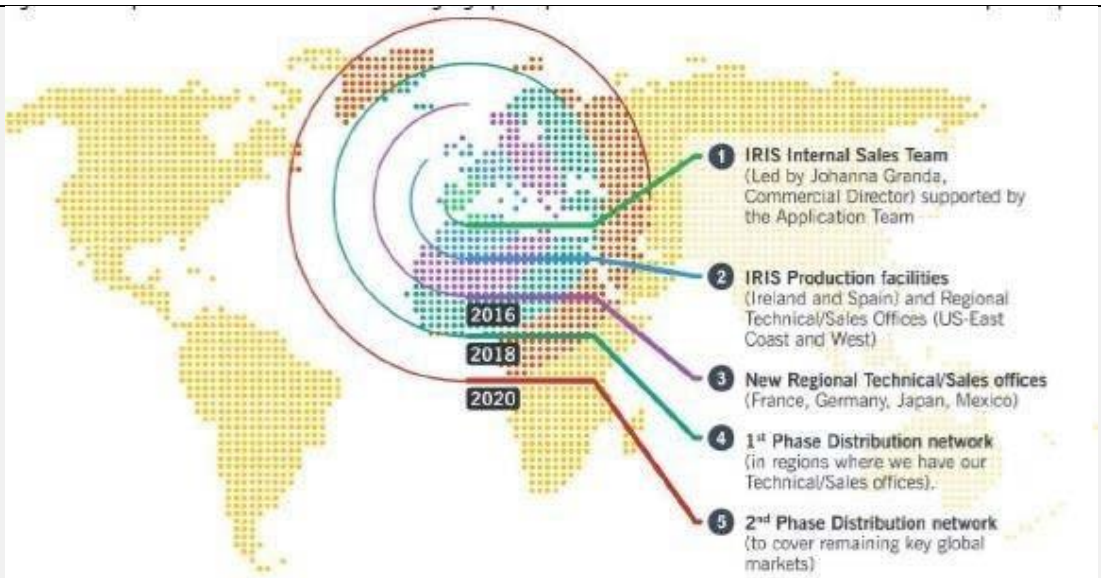
**DR. TIMOTHY KEHOE**. Male PhD in Engineering, University College Cork, Ireland- developed an IR photometric measurement system for thermal diffusivity in anisotropic materials. Masters in Optoelectronics, Queens University Belfast. Designed an Echelle spectrometer while based at Andor Technology Ltd. Worked at Tyndall National Institute, Cork, and Catalan Institute of Nanotechnology, Barcelona, developing diffractive optical and acousto-optical metrology techniques for nanotechnology applications, in particular for characterisation of nanoscale structures fabricated by nanoimprint lithography and directed self- assembly. Other research activities: quantification of ordering in self-assembled structures by image analysis techniques and optical sensors using hydrogel materials.

**OONAGH MC NERNEY** Female Director and Co-founder of IRIS. Hons. Degree in International Business (1992), Uni. of Stirling, Scotland. Prior to establishing IRIS in 2007, she spent 15 years working in the industry, technology institutes and Business Development Agencies in the areas of technology transfer, innovation, business development and industry foreign direct investment. Expertise in new product development, leadership of applied R&D, commercialisation, branding and marketing.

**COLM DIGBY.** Male. General Manager and Co-Founder of IRIS. B.Sc. Hons. in Applied Computing and a Diploma in Industrial Studies from the University of Ulster, Magee College, N.Ireland. Prior to IRIS he worked in the IT sector throughout Europe holding positions at IBM Europe in Paris, European Space Agency in Rome and Picas SA in Barcelona, where he was involved in many trans-European projects involving the sharing of data across international borders. With PICIS he has been involved in the implementation and support of PICIS Health Care system around Europe and the US.

**JOAN PUIG LAGO. MALE.** Operations Director at IRIS. Engineering Degree in Electronics from UniversitatPolitecnica de Mataróand in Industrial Organization from Escola UniversitatPolitécnicad'Enginyers Industrials de Terrassa. Executive Education inProgram for Management Development from ESADE. 12 years industry experience in automation of industrial processes for the main automotive & aeronautics companies: Mercedes, VW, SEAT, EADS, BOEING, etc. Solid knowledge in definition of strategies and targets, and aligning systems and workflows, according to the operations schedules and within agreed costs and high quality levels. Extensive experience in managing Technical Departments and Operations such as organization and monitoring of production and processes, development of operational plans, and giving guidance in the design and in the implementation of the final solutions

# IRIS SALES TEAM, DISTRIUTION CHANNELS AND PRODUCTION FACILITIES INFOGRAPHICS




**Participant Number**
**3**

| | |
|---|---|
| TWI | **Organisation:** TWI Ltd.<br>**Website:**www.twi-global.com |

**Organisation Description:**
TWI Ltd (TWI) works across all industry sectors with expertise in key aspects of materials, materials joining, structural integrity, static and dynamic testing and Non-Destructive Testing (NDT) and Non-Destructive Evaluation (NDE). The majority of TWI's activities (65%) are contracts, funded by industry and built around the development of advanced technologies. TWI is also an active participant in collaborative programs in Europe. TWI has a substantial fee paying industrial membership of around 3000 members from 70 countries centered mainly in energy-related business sectors such as oil & gas, petrochemical, process plant, power generation and their support industries (e.g. oil & gas plant maintenance and NDT). Also, in the transportation industry sector of aerospace/air-transport, marine/shipping, rail and automotive and their support manufacturing and maintenance organizations. This extensive network with the industry and its state-of-the-art facilities makes TWI a perfect partner for the **SmartSense** optimization & validation and commercialization process, creating a direct connection with international markets and targeted industries. TWI has a continued growth reaching €80M revenues and assets for over €100M in 2014. Currently with €8M of financial surplus and more than 1000 employees, TWI is one of the most successful RTO in the world. TWI has been involved in over 90 European framework projects, many of which have been in the field of NDT and Condition Monitoring involving ultrasonic testing. Projects of this type are usually targeted at the aerospace, oil & gas and energy related industries. TWI, therefore, has the ideal credentials to deliver a successful project on time and within budget. TWI will be heavily involved in SmartSense exploitation at the end of the project and particularly where the project developments are highly relevant to their industrial membership that is currently around 3000 organisations in over 70 countries.

**Role in the project:**

- Leading Project management as Coordinator of the SmartSense product optimization using TWI's expertise in product development, NDT and optimization.
- Risk management plan for both commercial and technical SmartSense (WP7)
- Customization of SmartSense for customer applications and adjustment of SmartSense system for precommercial application.

**Relevant Publications:**


- Lee Smith, Mike Gittos and Philip Threadgill, 1999. High quality and productivity joining processes and procedures for titanium risers and flowlines. Titanium Risers and Flowlines' Trondheim, Norway, 17 February 1999; published by SINTEF, Trondheim, Norway, 1999.
- J R Rudlin, C R A Schneider and G R Razmjoo, 2004. Reliability of inspection for root flaws in riser girth welds 23rd International Conference on Offshore Mechanics and Arctic Engineering OMAE, Vancouver, Canada 20-25 June 2004.
- Graham R Edwards, 2007. Detection of corrosion in offshore risers using guided ultrasonic waves. 26th International Conference on Offshore Mechanics and Arctic Engineering, OMAE 2007, San Diego, California, 10-15 June 2007. Paper no. 29407.
- Amir Bahrami, 2009. Fatigue performance of riser girth welds for deep-water applications. Deep Water Technology Conference, Asia, Kuala Lumpur, Asia.
- Colum M. Holtam (TWI North America LLC), David P. Baxter (Atkins Oil & Gas), 2001. Fatigue performance of sour deepwater riser welds: crack growth vs. endurance. ASME 30th International Conference on Ocean, Offshore and Arctic Engineering, OMAE 2011, Rotterdam, The Netherlands, June 19-24, 2011.

Relevant Products:

Tank integrity monitoring; non-invasive monitoring of storage tanks. The Tank Integrity Monitoring (TIM) project is working to develop a low-frequency ultrasonic technique for the non-invasive condition monitoring of tank floors to detect corrosion or cracking degradation.

Plastic pipe inspection made simple with PolyTest; a modular system consisting of wheeled chain links that are mounted on the outside of the pipe adjacent to the weld to be inspected. The chain link system is rotated around the circumference of the weld, enabling the attached transducer to complete a full volumetric examination.

PolyTest on-site testing

Infrastructure/Equipment:

Extensive laboratory facilities enable products to be assembled from very small geometry electronics and medical diagnostics devices to large power generation and cooling systems. TWI's national and international networks also allow it to source suitable suppliers of materials, machining, assembly, inspection and control equipment. It can also provide training and post production back-up in terms of the production line and product troubleshooting and additionally, it can assist in product upgrading and life extension.

- Full-scale pipe bend test rig
- Full-scale pressure and burst test facilities
- Seven resonance test rigs for full-scale accelerated fatigue testing of pipes
- Weld corrosion testing facilities
- Sweet and sour service testing capability at high temperatures and pressures
- High-pressure hydrogen testing facilities
- Small, medium and large-scale testing in H2S
- Testing facilities to supply NDT, mechanical and corrosion testing
- Rapid decompression testing
- Fluid compatibility testing

TWI facilities.

| Profile of Main Persons | |
|---|---|
| ***Name: Angela Angulo*** | ***Gender: Female*** |
| ***Role: Project coordinator*** | ***Education:BSc, MSc Structural Integrity*** |

| | |
|---|---|
| Ángela is a Condition Monitoring Project Leader at TWI. Ángela has a master's degree in Industrial Engineering from the University of Navarra, and she is pursuing further study in the technical area of Structural Integrity. She started her career development at Advanced Technologies Inspiralia, on the development of electronic control systems, mechanics, hardware and communications software development area. She was also working in collaboration with the business development strategy, validation of concepts, identification of partners and report development. On January 2013 Ángela joined TWI, Integrity Management Group, which provides costs savings and reliability assurance through the development and application of advanced innovative inspection, assessment and risk management solutions for Oil & Gas or any other industry. Within the Condition Monitoring Section, Ángela works on the following areas of expertise: SHM using vibration analysis, acoustic emission, modal analysis and guided ultrasonic waves, LRUT, novel ultrasonic techniques (emats, magnetostrictive transducers) and NDT performance demonstration. | |
| ***Name: Graham Edwards*** | ***Gender: Male*** |
| ***Role: WP2 leader*** | ***Education: C. Eng, BSc Hons.*** |
| Graham Edwards has over 40 years of experience in Non-Destructive Testing (NDT). Along with several other graduates, he started his career in NDT with the Unit Inspection Company, where he received training in all aspects of NDT and weld inspection. In 1973, he was one of the first ultrasonic test technicians to gain CSWIP approval. For the next four years, he worked in the N. Sea offshore oil, the French nuclear power and Middle Eastern petrochemical industries. In 1977, he joined Baugh and Weedon Ltd. as an Applications Engineer and was involved in the development of the award-winning 'Seaprobe' underwater ultrasonic test equipment. In 1981, he became a lecturer in the School of Applied NDT at The Welding Institute (TWI), before moving over to the NDT Research department in 1986. There, his research projects included NDT of plastics and composites, expert systems, and acceptance criteria in BS5500 for ultrasonic testing. He also set up the specialist site services team. While at TWI, he collaborated with the French Welding Institute on the NDT of fillet welds and was seconded to the Edison Welding Institute in the USA for two years to set up the NDT section there. | |

**Participant Number 4**

**Organisation Name: FloteksPlastikSanayiveTicaret A.Ş. (Floteks)**
**Website: www.floteks.com.tr**

**Organisation** Description:

FLOTEKS, since 1983, is a fully integrated Rotational Moulding production plant and is the first of its kind in Turkey. It is located in Bursa, the heart of Turkish Automobile Industry; 20 km from one of the biggest seaports in Turkey, Gemport. Floteks employs 17 engineers and 23 technicians within its 146 people staff in order to maintain leading technologies that enable customers around the world to build safer, lighter, durable plastic articles for vehicles, infrastructure, municipalities, FMCG, storage, gardens, homes, etc. Through memberships in many professional and trade associations, as well as extensive QC, research and development facilities, Floteks has been successful in supplying many complicated technical and difficult to produce products to the market. Plastic liner parts of some of the newly introduced composite fuel tanks in the market are products of Floteks. The company has certifications ISO 9001, EN 16949, EN 14001, 16949, 14001, EN 13589, EN 1176, OHSAS 18001001, obtained from organisations such as TÜV, BSI and TSE and a fully equipped Plastic Products R&D lab. For the last 8 years, Floteks has developed its own infrastructure of products using the TUBITAK framework and has successfully sold the know-how of these products to India (Vectus Inc., www.vectus.in ) and made a license agreement for 7 years.

- Production Pallett: 70 % of the production for automotive industry, main customers OEM's, Mercedes, Suzuki, CNH, Ford, Hyundai, Renault… Diesel tanks, Ad Blue tanks, air ducts, water tanks of various sizes, roofs being among the main products. Complying with all the product standards of the main automotive OEM's
- 30 % of the production various rotomolded articles, from 1 kg weight to 700 kg weight. Floteks is touching a wide variety of sectors. From water, chemical, rain harvesting tanks, for both underground and above ground storage, manholes for infrastructure, sewage systems, playgrounds, trash containers, both underground and above ground… Production complying with many EU standards e.g. EN 13598, EN 1176, etc. Customers

DAIMLER
MAN
RENAULT
Türk Traktör
CNH
ÇUKUROVA
Mercedes-Benz
Ford
FIAT
NEW HOLLAND
AGRICULTURE
HIDROMEK
EvoBus
KOLUMAN
TÜMOSAN
BMC
KARSAN
TEMSA
GLOBAL
HATTAT
AGCO
MOFFETT
TIRSAN
STILL
ERKUNT
SANAYI A.Ş.
AZIMUT
YACHTS
HERRMANN
IMER
GROUP
Putzmeister
JOHN DEERE
DALGAKIRAN
kompresör

Relevant Publications:

- “An ERP system for Rotomolding” by Mehmet Beysel, Rotoworld, Volume X Issue 4, 2014, P.56
- “Maintaining QC and QA as a Rotomolder” by Celal Beysel, Rotoworld, Jan / Feb 2015
- EU Funding for Rotomolders Facilitates R&D and Innovation by CelalBeysel, May / June 2015
- Various presentations with various topics by Celal Beysel throughout the world in Associations of Rotomolders, sinc years for example in Cairns/ Australia, Minneapolis / USA, Lyon / France, Ostend/ Denmark, New Delhi / India, Goa lately, the presentations in ARMO event Sept. 2015 in Nottingham, GB: “Can a Rotomolder Afford to Research and and in STAR India event in Goa, February 2016, “ Do You Know Your costs” and “NDT Systems for Rotomolders”.

Product Range:

Floteks has designed various products with IP rights, from playgrounds to tanks and also has many patent applications, Turkey and abroad.

- Both Vectus and PDL are acting as agents in their respective countries, Vectus already selling some products in
- Various distributors in Iraq and Saudi Arabia.

Watermeter & Checkvalve Chambers – Pump Stations

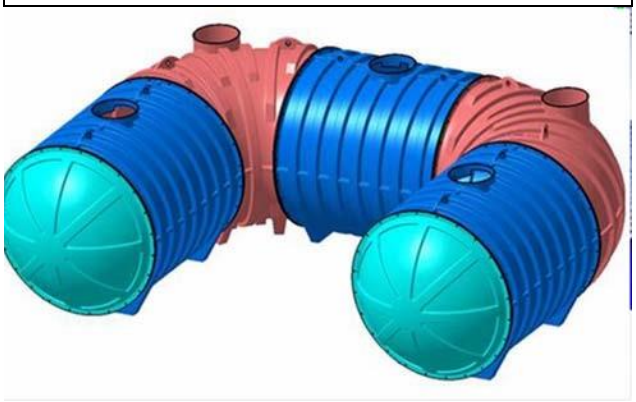
DN 400 SERIES

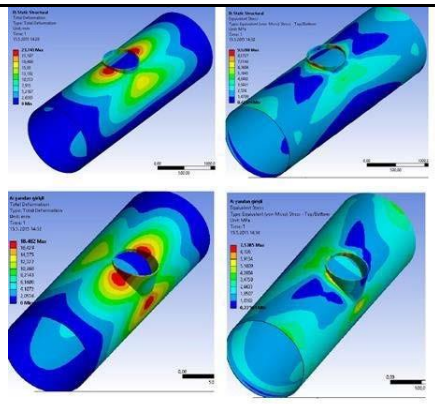
PE UNDERGROUND TANKS

PE 5.000 LITERS

Underground PE Waste Systems

**Underground container**





## Patent applications

| Subject | Name | Drawing/Photo | Country | Application Date | Nr. | Referance Nr. | Situation |
|---|---|---|---|---|---|---|---|
| PATENT APPLICATION | Heat Pipe application in Rotomolding | | TURKEY | 13/10/2015 | 15/12708 | TP15-185 | report expected |
| PATENT APPLICATION | Leakage Control | | TURKEY | 25/06/2015 | 15/07826 | TP15-183 | report expected |
| PATENT SEARCH | Leakage Control | | Worldwide | | | TP15-187 | completed |
| PATENT SEARCH | HEAT PIPE | | Worldwide | | | TP15-184 | completed |
| Design right | Corrugated Pipe | | TURKEY | 11/09/2015 | 15/11348 | TF15-023 | under investigation |
| PATENT APPLICATION | Metal Insert for Tanks | | TURKEY | 20/11/2014 | 14/13730 | TP14-181 | report expected |
| PATENT APPLICATION | Modular Cow housing | | TURKEY | 09/01/2015 | 15/00250 | TP14-173 | report expected |
| Design right | Metal inlet system | | TURKEY | 15/10/2014 | 14/12111 | TF14-023 | certificate expected |
| PATENT APPLICATION | Multi Purpose plastic column holding modular walls | | TURKEY | 10/09/2014 | 14/10624 | TP14-193 | report expected |
| Design right | Special Pallett design for 19 lt water bottles | | TURKEY | 27/03/2013 | 13/03721 | TF13-043 | Certificated |
| Design right | Welding machine | | TURKEY | 11/10/2012 | 12/11701 | TF12-050 | Certificated |
| Design right | Hygenic Container | | TURKEY | 12/05/2009 | 09/03682 | TF09-019 | Certificated |
| Design right | Water meter chamber | | TURKEY | 05/02/2010 | 10/00887 | TF10-018 | Certificated |
| Design right | Platic Menhole | | TURKEY | 11/02/2011 | 11/01340 | TF11-037 | Certificated |
| Design right | Platic Menhole lid | | TURKEY | 24/08/2011 | 11/08544 | TF11-038 | Certificated |
| Design right | Lid design | | TURKEY | 26/08/2011 | 11/08658 | TF11-039 | Certificated |
| Design right | Welding and drilling device for Manholes | | TURKEY | 09/10/2012 | 12/11558 | TF12-049 | Certificated |
| PATENT APPLICATION | Underground trash container | | TURKEY | 10/10/2014 | 14/11907 | TP14-132 | certificate expected |
| PATENT APPLICATION | Portable Mould cooler | | TURKEY | 20/03/2014 | 14/03290 | TP14-016 | certificate expected |
| Design right | Underground trash container housing | | TURKEY | 17/01/2014 | 14/00567 | TF14-001 | certificated |
| PATENT APPLICATION | Metal Baffle for plastic fuel tank | | TURKEY | 09/01/2014 | 14/00261 | TP13-183 | report expected |
| PATENT SEARCH | Portable Mould cooler | | Worldwide | | | TPD13-017 | completed |
| PATENT SEARCH | Forklift clamping device | | Worldwide | | | TPD13-001 | completed |
| PATENT APPLICATION | Portable water cooler | | TURKEY | 31/01/2013 | 13/01245 | TP12-141 | Certificate expected |
| PATENT APPLICATION | Portable water cooler additional patent | | TURKEY | 20/06/2013 | 13/07452 | TP13-030 | Report expected |
| PATENT APPLICATION | Metal Baffle for plastic fuel tank | | INDIA | 26/11/2014 | 3761/MUM/2014 | YP14-028/IN | Under investigation |
| PATENT APPLICATION | Underground plastic container housing | | WIPO | 19/01/2015 | PCT/TR2015/000027 | YP15-001 | report compleded |
| PATENT APPLICATION | Metal lid insert for plastic tank | | EPC | 15/10/2014 | 15169366.0 | YP15-015/EP | Under investigation |
| PATENT APPLICATION | Metal lid insert for plastic tank | | INDIA | 23/04/2015 | 1647/MUM/2015 | YP15-015-IN | Under invesigation |

**Some services that Floteks provides:**

**PRODUCT TESTS**

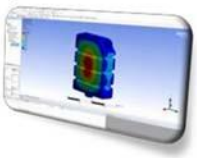
**Regulation tests**

CONSTRUCTION & DEVELOPMENT

SOFTWARE

| | |
|---|---|
| Catia V5 | Design Engineering & Mold machining |
| Visi-Cad | Mold design |
| Visi-Cam | Manufacturing |
| Alias Studio Tools | 3D Design |
| Studio Max | 3D Design |
| ANSYS Analysis | FEA |

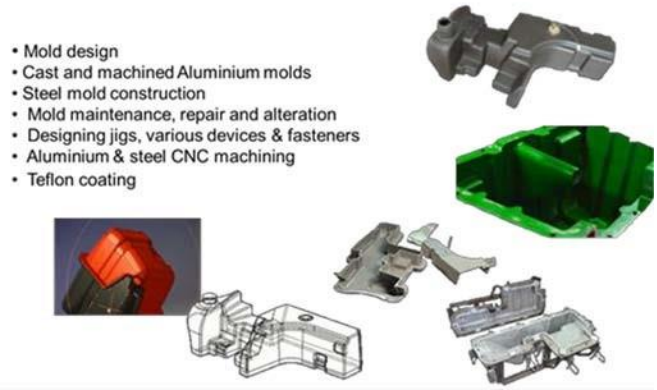
- Design and construction support
- Product design & co-design
- Metal to plastic conversion
- Finite Element Analysis
- Prototype development
- Modeling
- Computer Simulation of Rotomolding



**CONSTRUCTION & DEVELOPMENT**

**Moulds**

- Mold design
- Cast and machined Aluminium molds
- Steel mold construction
- Mold maintenance, repair and alteration
- Designing jigs, various devices & fasteners
- Aluminium & steel CNC machining
- Teflon coating



**Relevant Projects:**

- Floteks has completed 5 different R&D projects using TUBITAK/Turkey funds successfully. Utilising the knowledge attained from one of the TUBITAK Projects, links have been established with Vectus Ltd in India for manholes and a licensing agreement with the company for 7 years has been created. Total Commercial Value of know how transfer, licence and mould sales to Vectus: over €900,000.
- Research for a Rotomoldable PP developed by an Australian company, PSD (www.plasticsales.com.au).As a result of this project, inspecting mould ability, weldability, low temperature and long term creep behaviour of the PP have developed further, with funding provided by PSD. Prof. Yusuf Menceloglu (yusufm@sabanciuniv.edu ) from Sabanci University in Turkey is the advisor in this project. Commercial value up-to-now for the PP research: over €100,000.
- Manufacturing trial productions of LFRT (Long Fiber Reinforced Thermoplastics) with twin-screw extruder, using PP and testing the effects of LFRT in Rotomoulded products, self funded.
- Participated in various EU Projects such as: Partner in F7 Project DeiceUT (concluded in August 2015) and Coordinator SME partner in FP7 “SafeHPower” project ( to be concluded in February 2016).
- Successfully submitted the H2020 SME-I Project “LeakFree” as the sole beneficiary and Coordinator. The Project started in June 2015. The aim is to detect leakage of tanks for OEM’s after production, using acoustic emission.

Infrastructure/Equipment:

Floteks has a production capacity of up to 4000 tons of plastic per year and can produce parts from 350 grams to 700 kilograms. With a separate R&D section which includes a modern laboratory as well as a quality control lab and experienced engineers to use the facilities.

- A Quality Control lab able to perform daily checks on the quality of plastic production, with a CMM facility to check dimensions of products and compare within the given tolerances, drawing among other necessary QC devices. The lab also has full capability of standard regulation testing of plastic vehicle parts according to EC – R 34 regulations.
- A special R&D lab, not involved in daily work, for material testing, with facilities for UV testing ( Q-Control ) , DSC, Climate Chamber ( -50 to + 150 Degrees C ) , MFI, Tensile strength testing ( 10 kN), Polarisation Microscope, Impact testing, Colour testing, Ultrasound Thickness testing and Density testing.
- Software for Product and Mould Drawing and FEA ( 5 x Catia, 1 x Unigraphics, 1x ANSYS ) plus necessary workstations and educated personnel to use these IT software.
- Spin welding, Hot plate welding, Extrusion welding, Hot Gas welding, Butt welding, also highly technical home- made semi- automatic welding equipment.

| Profile of Main Persons |
|---|
| ***Name:V.CelalBeysel***<br>***Gender: Male*** |
| Background:<br>CelalBeysel founded Floteks in 1977, as a small company with 4 personnel. He has developed the company to be the leading rotomolder, known worldwide. Besides being an entrepreneur, he has been chairman of major business associations both local and nationwide (4 years BUSIAD and 5 years TURKONFED). Celal has worked as the president of a respected foundation for Vocational Education in Turkey (5 years BEGEV), having written many articles on Vocational Education, SME's, R&D and Innovation, published in Turkish newspapers and magazines in Turkey and abroad.<br><br>Education:<br>Chemical Engineer graduated from Zurich Technical University ETH, speaks English, German and French.<br><br>Work Experience:<br>President of the Board and Founder of Floteks, a major plastics manufacturer in Turkey using Rotational Moulding Technology.<br><br>Product Development Experience:<br>• Constructed his own rotomolding machines in 1983, which the company used till 2000, until he decided to go over to state of the art German designed machinery<br>• Personally constructed pulverisers used for rotomolding production in 1983, which are still in use, after some renovations.<br>• Constructed the extruders used in the company for compounding, making renovations when necessary.<br>• Has led the design team to design products and moulds for production in Floteks, since the beginning.<br>• All product development, R&D and Innovation efforts for products, raw materials and finishing jigs/devices in Floteks are lead, advised and controlled by CelalBeysel.<br>• Currently leads the automation efforts in the company, for welding and NDT mostly.<br>• Has developed the technical knowledge for manholes and infrastructure products for Floteks.<br>• Has introduced the know-how of manholes to India and sold knowledge/ license to a major company, with a commercial value of €900,000.<br>• Has developed underground garbage collection system for a municipality in Bursa, achieved a sales value of €600,000.<br>• Has bought up the idea for the EU project "Leakfree" and helped in writing the project proposal.<br>• Has led the PP development and testing efforts for Australian company PSD, sales value over €100,000.<br><br>Intellectual Property Management:<br>CelalBeysel is leading the IP rights applications and management in the company. The IP Protection in Floteks started in the 1990's, when Floteks introduced Rotomoulded Playgrounds to the Turkish market, but was severely copied. Starting in 2000, when the company managed the "Quantum Jump" to be a Co-designer for Automotive OEM's, the competition started to copy technical articles as well. CelalBeysel has been instrumental in defending Floteks' rights with patents.<br><br>Innovation Management:<br>Floteks, being a supplier to the automotive OEM's, has to stay a technological leader and innovation / R&D champion, to be able to protect its position as market and technology leader within the very competitive automotive world. Therefore, CelalBeysel, as the founder of the company, has left the daily concerns of the company to the experienced Floteks team and is occupied with developments needed for the future of Floteks. |

CelalBeysel is known to the rotomolding world, as he is attending many meetings of the rotomolding associations throughout the

world, for networking and also presenting his ideas / know how / leadership. This way, he was able to commercialise the manhole know how of the company, achieved by doing a research project with the Turkish funds of the organisation TUBITAK, also persuade the Australian PP developer to use Floteks' premises and knowhow for testing the material. Receiving SME-I Phase 2 Funding for the Project LEAKFREE for his company FLOTEKS was his last success.

***Name: Ali Balcı***
***Gender: Male***

Background:
Has knowledge in the use of Catia V5R16, Autocad, Ansys FEA, Solidworks. He has training and Certificates for Control Apparatus Design, Design Seminar on Rotational Moulding Technology, Metrolog XG VER 9.002, Geomagic Studio, Patran Training, Statistical Process Control, A.P.Q.P. an P.P.A.P. , ISO/TS 16949:2002, MSC M arc – Patran Nonlinear FEA Training, ANSYS Nonlinear Training, Training in Efficiency and Cost Reduction Techniques, CNC Milling, CastAluminum and Sheet Metal Mold Design and Production Training. He is the most experienced mechanical engineer in the company, having an insight in CAD, CAM, FEA, mold design, product design, production and engineering analysis.

Education:
Mechanical Engineer graduated from Uludağ University Bursa (2004), speaks English. Attended many courses in CATIA, ANSYS,
ROBOTICS, Mitsubishi and Siemens PLC's, Festo Academy.

Work Experience
2 Years AKTAS Holding, a major EPDM compounder and producer for Air Springs for
Vehicles in Turkey 10 years Floteks, a major Plastics manufacturer in Turkey using
Rotational Moulding Technology

Product Development Experience:
In charge of the R&D Lab in Floteks. With his education, training and long years of experience, he is designing and producing automation devices for Floteks, also programming the PLC's assembled in these machines.

Intellectual Property Management:
Ali is one of the major contributors within the firm for writing patents; additionally he is responsible for creating the designs of the products to be patented.

Innovation Management:
Leads mechanical innovation and R&D activities, makes FEA's for new products.

***Name: Mehmet Beysel***
***Gender: Male***

Background:
Started life as an entrepreneur, Mehmet gained significant international experience selling the products of his company Flopark throughout the world, including: Sudan, USA, GB, Germany, Russia and Israel. Later, he handed Flopark over to a professional team, however he stayed on as the president of the board whilst taking over the daily running of Floteks. He speaks English and
German.

Education:
Graduated in 2000 from Duke University, Durham / USA, Major in International relations, Minor in Economics

Work Experience:
Took over the company, Flopark, a Playground production company, in 2001. The company grew to make a turnover of €2 million per year, selling Playgrounds in Turkey and exporting. In 2010, he became the Automotive Sales Director at Floteks, before becoming the General Manager in 2013. Through his work in making the automotive operation more productive and competitive, he was able to expand Floteks' automotive business and export products to a variety of countries. His success in introducing Floteks to Hyundai, will double the turnover of Floteks per year within 2017. Floteks achieved the Innovative Company of the Year award under his management.

Product Development Experience:
The new products he has introduced to Floteks will show their effects in 2017. He was able to double the orders from Mercedes Benz, which will reach €1,000,000 turnover in 2017. Furthermore, he persuaded Hyundai to choose Floteks as their Adblue tank provider starting in 2015, reaching a turnover of €7 million per year in 2017. The Leakfree project was initiated with the future need of doubling the leakage control of the tanks produced in Floteks.

Innovation Management:
The Hyundai tank was actually a blow moulded tank. It needed some changes in design, to make the tank with rotomolding technology. It has cost Floteks two visits to the Hyundai technical team in South Korea to persuade them to change over to rotomolding and also a lot of development efforts between both parties.

***Name: Canan ERAT***
***Gender: Female***

Education:
Chemistry graduated in 1994 from Gemlik High School Bursa, speaks English. Attended many courses in Microsoft office, Quality Management Systems and many plastics test applications.

Work Experience:
5 years Floteks, a major Plastics manufacturer in Turkey using Rotational Moulding Technology, as a technician. She is using laboratory equipment of Floteks, to make the required daily tests. DSC, Tensile Strength, QUV, Drop and Impact Test, Density, MFI, Ring Stiffness, Vacuum Test devices, polarised Microscope and Ultrasound thickness Gauge are the devices she uses. She has been working on the SME-I Project LEAKFREE since she started working in Floteks.

Product Development Experience: She has an experience for testing
plastic products using the necessary equipment..

Innovation Management:
She works on R&D activities and product development as a laboratory technician.

**Name: EmreAkbay**
**Gender: Male**

Education:
Automotive Engineer graduated in 2012 from Marmara University Istanbul , speaks English. Attended many courses in Solidworks, Catia, ANSYS.

Background:
He has training in CAM and engineering analyses. He is fluent to use Catia V5R16, Solidworks and ANSYS.

He is also using laboratory equipment: DSC, Tensile Strength, QUV, Drop and Impact Test, Density, MFI, Ring Stiffness-Control the Temperature monitoring, Vacuum Test devices.

Work Experience:

4 years at Floteks, a major Plastics manufacturer in Turkey using Rotational Moulding Technology. He worked 2 years as a manufacturing process engineer .He has been working as a research engineer since 2014.He has been working on the SME-I Project LEAKFREE since he started working in Floteks.

Product Development Experience:

Technical Designer and Research Engineer in the R&D Department of Floteks.

He is also designing new underground products and fixtures and making the FEA's. He has an experience for testing plastic products.

He is also defining process parameters of new products and improvement existing products using temperature monitoring system.

**Participant Number 5**

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|---|---|
| Brunel UNIVERSITY LONDON BIC | **Organisation Name: Brunel University London (BUL)**<br><br>**Website: www.brunel.ac.uk/BUL** |

**Organisation Description:**

Brunel University London has been highly successful in creating an industrial innovation research resource, named Brunel Innovation Centre (BUL), which sits between the knowledge base and industry, supporting partners in industry to transfer academic research into industrial application. Brunel has a wide range of knowledge and expertise in the following research areas: (i) Power ultrasonics; detection, cleaning of fouling and fouling prevention techniques non-invasively software (ii) Smart NDT; autonomous devices, miniaturised multifunction systems for NDE in bespoke environments; marine, high temperature, aerospace, pattern recognition hardware / software, (iii) Active SHM and Condition Monitoring (iv) Design of associated hardware and software (v) Signal and image processing and numerical modelling - Noise reduction, signal condition and control (including active vibration control), data interpretation and analysis, (vi) Automation and robotics of inspection system, (vii) Improved or new and robust sensors (Piezoelectric, Electromagnetic) e.g. high temperature inspection, transduction (viii) Electronics - Low power consumption for multiple channels, wireless, high speed, wideband, fast bps, energy harvesting. In year 2015, BUL has received the Highly Commended Innovation Awards from The Institution of Engineering and Technology (IET) for two award categories: (i) Measurement in Action and (ii) Manufacturing Technology. The collaborative European project (FP7) was put forward to the competition and was chosen from over 300 entries from 28 countries (globally).

IET AWARDS The Institution of Engineering and Technology | INNOVATION FINALIST 2015

**Role in the Project:**

BUL will lead the Certification, Quality and Maintenance, and Training Development work package and produce training manuals and product documentation. BUL will bring their expertise in developing complete, concise and organised documentation for best practices. Relevant training material including Getting started section, user guide, examples, screen cast tutorials and FAQs will also be embedded in Help section of the software itself. BUL will coordinate the Awareness and Promotion events as part of the Dissemination, Exploitation and Communication work package. This will include interaction with end-users in power sector through newsletters and press releases. BUL will assist TWI in coordinating the project by managing financial, regulatory and operational risks in the service of innovation by analysing statistics, reports and market trends.

**Main Partners & Collaborations:**

**Relevant Projects:**

- Adapted Composite Repair Tooling for in-situ wind turbine blades structural rehabilitation (CORETO), funded by FP7. Project website: http://coreto-project.eu/index.php
- Automated Ultrasonic Inspection of Aerospace Composites with Enhanced Defect Detection Probabilities Aided by Gantry Deployed, CAD Controlled Robotics (AutoDISC), funded by Innovate UK. Project website: http://autodisc-project.com/
- Radio Frequency Sensing for Non-Destructive Testing of Carbon Fibre Reinforced Composite Materials for Structural Health Monitoring (Comp-Health), funded by FP7. Project website:http://comphealth.eu/
- An ICT Enabled Approach to Optimising the Reliability of Manual Ultrasonic Non-Destructive Testing (ICARUS), funded by FP7. Project website:http://www.icarus-project.eu/home/index.jsp
- Continuous monitoring systems for the SAFE storage, distribution and usage of Hydrogen Power for transport

**Infrastructure/Equipment:**

- Management, administration and accounting department
- Design and development of signal processing and pattern recognition software
- Data management and handling software
- NDT and Condition Monitoring hardware i.e. Acoustic Emission and Ultrasound equipment
- Power amplifier, oscilloscope and impedance analyser
- Optical tracking device and High resolution cameras
- Several powerful work-stations including signal and image processing software
- State of the art 3D Laser scanning vibrometer for: wave propagation; transducer analysis; ultrasonic detection / cleaning; ultrasonic anti-fouling; condition monitoring purposes

Experimental industry specific set-up rigs for detection and cleaning of fouling, rotating machinery; mock-ups; inspection, monitoring systems demonstrated in full scale.

**Profile of Main Persons**

**Name*:* Dr Alvin Chong**
**Gender**: Male

**Qualifications:** BEng (Hons), PhD, CEng, IntPE (UK), MIET, MIOA, MIEEE

**Experience/ Career History:** Dr Alvin Chong graduated with a BEng (Hons) degree from University of Hull (UK) in Electronic Engineering. He holds a PhD from Open University (UK) where he researched on acoustic wave propagation in periodic structures (sonic crystal). He joined BUL in March 2012 and currently he holds the responsibility as a Research Fellow Team Leader for the Smart Non-Destructive Testing (NDT) team. His research in BUL predominately focused on the novel NDT methods for the aerospace applications.

**Work Experience:** Dr Alvin Chong undertook the roles as Project Coordinator and Project Leader for various European Commission (EC) Seventh Framework Program (FP7), Clean Sky and Innovate UK research projects. Many of which have either been successfully completed or research in progress.

**Product Development Experience:** In light of his research activities in BUL, he has developed various functional prototypes demonstrated to mid-tier Technology Readiness Level (i.e. TRL 4 to 6).

**Innovation Management:** He has published more than 25 papers in international refereed journals and conferences on ultrasonic NDT, electromagnetic NDT, laser system, acoustic band gap material, structural analysis, local resonance material and computational modelling based on finite element method.

**Name*:*** Liang Chen
**Gender:** Male

**Qualifications:** BSc, MSc, PhD, MIEEE

**Experience/ Career History:** Dr Liang Cheng received the B.Sc. degree in electronics from Peking University, Beijing, China, in 2007, the M.Sc. degree in communications and signal processing from Imperial College London, London, U.K., in 2008and the Ph.D. degree in failure models and life cycle assessment of wind turbine systems at Newcastle University, U.K in 2013.
He has involved several EPSRC and FP7 projects for development of electromagnetic, thermographic and ultrasonic NDT systems. In 2013, He received PhD degree from Newcastle University, UK. His research topic is non-destructive evaluation for carbon fibre reinforced plastic wind turbine blades using eddy current pulsed thermography (ECPT). During the PhD study, part-time RA at Newcastle and the exchange in China, Dr Cheng developed and re-designed the ECPT sensing system and was also involved design of eddy current and magnetic flux leakage system with colleagues. He developed ECPT technique for both composite and metallic components for the detection and quantification of crack, corrosion, delamination and impact damage through both simulation and experiment. Signal/image processing algorithms were developed using Matlab for feature extraction and defect detection, classification, quantification. From 2013, he joined Brunel Innovation Centre. His current research interests cover electromagnetic acoustic transducer design for absorb tubes used in concentrating solar power, guided wave transducer for de-icing of wind turbine blades, guided wave modelling for complex geometric structure such as power cables, nuclear piping etc.

**Work Experience:** Dr Liang Cheng undertook the roles as Project Leader or got actively involved in European Commission (EC) Seventh Framework Program (FP7) and Innovate UK research projects. Many of which have either been successfully completed or research in progress.

**Product Development Experience:** In light of his research activities in BUL, he has developed electromagnetic acoustic transducer prototypes for monitoring of absorber tubes in solar plants at high temperature, eddy current pulsed thermography NDT&E system for composites and metallic structures, phased array on time-of-flight-diffraction configuration for high temperature pipes, demonstrated to mid-tier Technology Readiness Level (i.e. TRL 4 to 6).

**Name:** Jiangtao Sun
**Gender:** Male

**Qualifications:** BSc, MSc, PhD, MIEEE

**Experience/ Career History:** Dr Jiangtao Sun received his BSc and MSc degrees in Measurement Technology and Instrumentation from Xidian University (China) in 2008 and 2010, respectively. His PhD was awarded in Electrical and Electronic Engineering from University of Manchester (UK) in 2014, specialized in electrical tomography for industrial applications. In September 2014, he joined BUL as a research fellow in Smart Non-Destructive Testing (NDT) team. He is currently working on automated defect recognition based on X-ray radiography, X-ray CT and ultrasonic C-Scan imaging. Dr Jiangtao Sun is currently involved in several Innovate UK projects as technical leader, including “automated in-line inspection and quality control of net shape powder metallurgy components using microfocus three dimensional x ray computed tomography imaging (Qualinet)”, “automated,100% production quality assurance of net shape manufactured components using in line micron resolution x-ray sterographic imaging”, and “automated ultrasonic inspection of aerospace composites with enhanced defect detection probabilities aided by gantry deployed, CAD controlled robotics (AutoDISC)”.

**Product Development Experience:** He successfully developed an embedded system for the locking of a second-class frequency standard to the reference signal from GPS, including hardware and software design, during his master study. He is currently developing an image processing toolbox suitable for industrial X-ray radiographic and CT imaging using Matlab.

**Innovation Management:** He has published more than 10 papers in peer-reviewed international journals and conferences, mainly focusing on electrical tomography sensor design and image reconstruction for industrial applications and image processing for non- destructive testing and evaluation using tomographic methods.

**Name:** Dr Haitao Zheng
**Gender:**Male

**Experience/ Career History:** Dr Haitao Zheng studied Engineering Mechanics as an undergraduate and postgraduate at Tongji University, Shanghai, China. During his doctoral study at Keio University, Yokohama, Japan, his research focused on the application of system identification and time series analysis to structural health monitoring. He had significant work and research experiences on structural health monitoring, NDT and condition monitoring, monitoring system implementation, and transforming research results to viable algorithms and application software.

**Work Experience**: Dr Haitao Zheng undertook the roles as Project Leader for various European Commission (EC) Seventh Framework Program (FP7) and Innovate UK research projects. Many of which have either been successfully completed or research in progress.

**Product Development Experience:** In light of his research activities in BUL, he has developed various functional prototypes demonstrated to mid-tier Technology Readiness Level (i.e. TRL 4 to 6).

**Innovation Management:** He has 22 published papers in international refereed journals and conferences on structural health monitoring, structural dynamic analysis, reflection seismology, wave propagation in structures, novel NDT in civil engineering etc.

**Name:** Maria Tyler
**Gender:** Female

**Experience/ Career History:** Maria received a BCom (Hons) degree in Accounting and Finance from the University of Birmingham. She joined Brunel Innovation Centre (BUL) in May 2014 as Project Coordinator. She has experience in portfolio management, client projects, contract negotiation and project management. She is responsible for the effective administration and coordination of BUL.

## 4.2. Third parties involved in the action (including use of third party resources)

| | |
|---|---|
| Does the participant plan to subcontract certain tasks (please note that core tasks of the action should not be sub-contracted) | Yes |
| The common view of the consortium following various discussions is that there is considerable room for improving the end product.<br>More specifically:<br>• The design of the 3-channel boards including wiring should be reviewed by an experienced electronics company whose core business is equipment development and manufacturing (Task 1.1). • Firmware should be improved in order to increase the speed of signal post-processing (Task 1.1).<br>• Software, Graphical User Interface should also be reviewed and improved in the light of discussions with potential end-users, (Kadoil and Shell being development partners in this project) (Task 1.2).<br>• The overall product structure needs to be optimized considering also cooling of the equipment (Task 1.1).<br>• The product should be packaged professionally with due consideration differing operational conditions. Ideally, different models of the product should be made available to the market depending on specific operational needs (Task 1.1).<br>Therefore, this work scope mainly related to WP1 will be subcontracted to Arlentus Control and it is expected from Arlentus Control that the first working prototype will be delivered to the Consortium to be tried during the site trials in Bursa and Ankara.<br>It was confirmed that the cost of subcontracting will be 20,000 EUROS.<br>The consortium has gone through the Best Value for Money (BVM) procedure and collated three proposals. Arlentus proposal was the proposal chosen by the consortium to be the best proposal for value to the expected outcome. On the 8th May 2018, it was agreed by a unanimous decision that the total value of subcontracting will be equally shared by the five members of the consortium. | |
| Does the participant envisage that part of its work is performed by linked third parties | N |
| Does the participant envisage the use of contributions in kind provided by third parties (Articles 11 and 12 of the General Model Grant Agreement) | N |
| Does the participant envisage that part of the work is performed by International Partners [2] (Article 14a of the General Model Grant Agreement)? | N |